The documentation and process conversion measures necessary to comply with this revision shall be completed by 24 September 1994

INCH-POUND

MIL-S-19500/511B
24 June 1994
SUPERSEDING
MIL-S-19500/511A
11 July 1987

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON,
SWITCHING TYPE 2N4261
JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for PNP, silicon, switching transistors. Four levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1 (similar to TO-72) and (see 3.3).

1.3 Maximum ratings.

P_T 1/ T_A = +25°C	V_{CBO}	V_{CEO}	V_{EBO}	I_C	T_J	T_J and T_{STG}	$R_{\theta JC}$
mW	V dc	V dc	V dc	mA dc	°C	°C	°C/mW
200	15	15	4.5	30	200	-65 to +200	.860

1/ Derate linearly 2.3 mW/°C above T_A = +25°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ELDT, 1507 Wilmington Pike, Dayton, OH 45444-5765, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

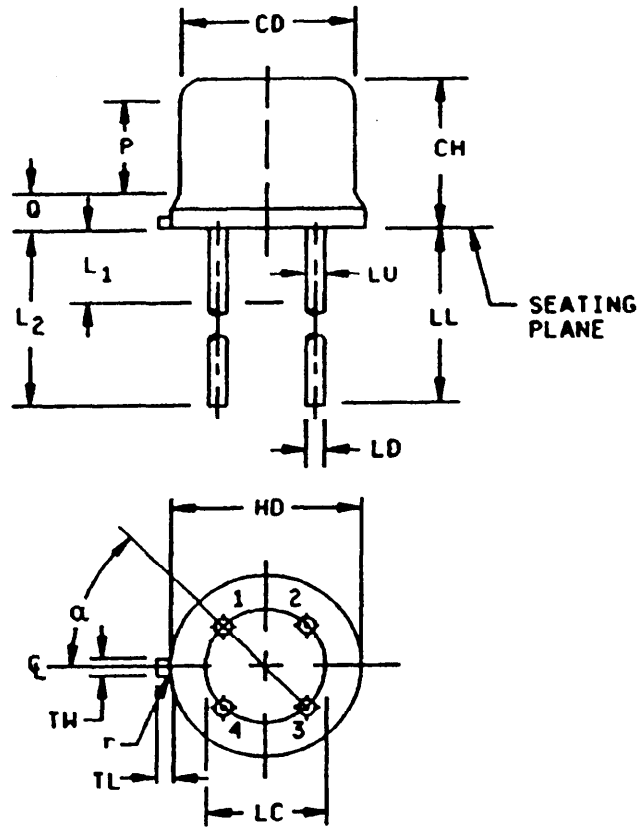
AMSC N/A

FSC 5961

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CH	.170	.210	4.32	5.33	
LC	.100 TP		2.54 TP		6
LD	.016	.021	.41	.53	7, 8
LU	.016	.019	.41	.48	7, 8
HD	.209	.230	5.31	5.84	
CD	.178	.195	4.52	4.95	5
TW	.036	.046	.91	1.17	2
TL	.028	.048	.71	1.22	3
LL	.500	.750	12.70	19.05	7, 8
L_1		.050		1.27	7, 8
L_2	.250		6.35		7, 8
P	.100		2.54		5
Q		.04		1.02	
r		.007		.18	10
α	45° TP				6



NOTES:
1. Metric equivalents are given for general information only and are based on 1 inch = 25.4 mm.
2. Beyond r (radius) maximum, TW shall be held for a minimum length of .11 inch (.028 mm).
3. TL measured from maximum HD.
4. Outline in this zone is not controlled.
5. CD shall not vary more than .010 inch (.25 mm) in zone P. This zone is controlled for automatic handling.
6. Leads at gauge plane .054 +.001 -.000 inch (1.37 +.03 -.00 mm) below seating plane shall be within .007 inch (.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by the gauge and gauging procedures shown on figure 2.
7. LU applies between L_1 and L_2. LD applies between L_2 and LL minimum.
8. All three leads.
9. This lead shall be electrically connected to the case.
10. r (radius) applies to both inside corners of tab.
11. Dimensions are in inches.

FIGURE 1. Physical dimensions TO-72).

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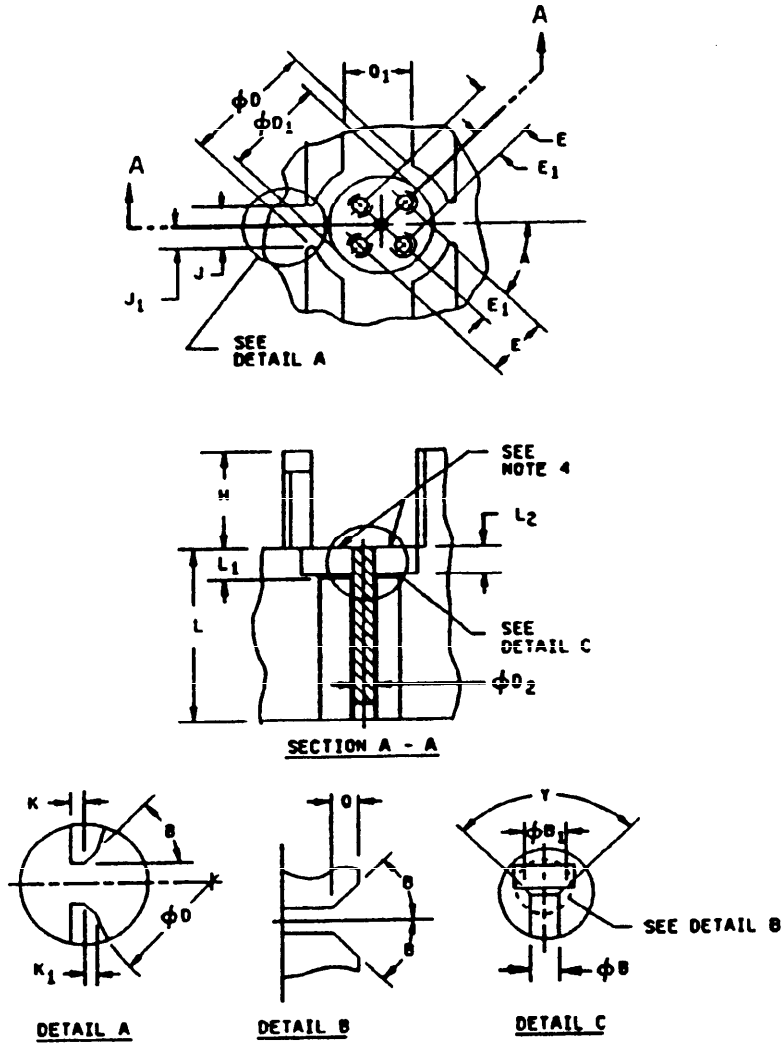


FIGURE 2. Gauge for lead and tab location.

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
ϕB	.0325	.0335	.83	.85	5
ϕB_1	.043 Nom		1.09 Nom		5
ϕD	.2310	.2315	5.87	5.88	
ϕD_1	.159	.161	4.04	4.09	
ϕD_2	.040 Nom		1.02 Nom		6
E	.0995	.1005	2.53	2.55	
E_1	.0495	.0505	1.26	1.28	
H	.145	.155	3.68	3.94	
J	.0470	.0475	1.19	1.21	
J_1	.0235	.0245	.60	.62	
K	.009	.011	.23	.28	
K_1	.005 Nom		.127 Nom		
L	.372	.378	9.45	9.60	
L_1	.054	.055	1.37	1.40	
L_2	.043 Nom		1.09 Nom		
Q	.040 Nom		1.02 Nom		
Q_1	.123	.127	3.12	3.23	
α	44.90°(min)		45.10°(max)		
B	45° Nom				
Y	90° Nom				

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The following gauging procedures shall be used: The device being measured shall be inserted until its seating plane is .125 inch (3.18 mm) +.010 inch (0.254 mm) from the seating surface of the gauge. A force of 8 ±.5 oz. shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gauge. The use of a pin straightener prior to insertion in the gauge is permissible. A spacer may be used to obtain the .125 inch (3.18 mm) distance from the gauge seat prior to force application.
4. These surfaces to be parallel and in same plane within ±.001 inch (0.025 mm).
5. Four holes.
6. Pressed in.

FIGURE 2. Gauge for lead and tab location.

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1.4 Primary electrical characteristics at T_A = +25°C, unless otherwise indicated.

	h_{FE} at V_{CE} = 1.0 V dc			$\|h_{fe}\|$	r'_b, C_{cl}	Pulse response	
	I_C = 1.0 mA dc	I_C = 10 mA dc	I_C = 30 mA dc	V_{CE} = 10 V dc, f = 100 MHz, I_C = 10 mA dc	V_{CE} = 4.0 V dc, f = 31.8 MHz, I_C = 5 mA dc	t_{on}	t_{off}
					ps	ns	ns
Min	25	30	20	20			
Max		150			60	2.5	3.5

Limit	$V_{CE(sat)}$		V_{BE}		C_{obo}
	I_C = 1 mA dc, I_B = 0.1 mA dc	I_C = 10 mA dc, I_B = 1.0 mA dc	I_C = 1 mA dc, V_{CE} = 1 V dc	I_C = 10 mA dc, V_{CE} = 1 V dc	V_{CB} = 4 V dc, I_E = 0, 100 kHz ≤ f ≤ 1 MHz
	V dc	V dc	V dc	V dc	pF
Min					
Max	0.15	0.35	0.8	1.0	2.5

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500, and as follows:

I_{BEX} . . . . . Breakdown voltage, emitter to collector, with base open-circuited.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and on figure 1 (TO-72) herein.

3.3.1 Lead finish. Where a choice of lead finish is desired, it shall be specified in the acquisition requirements (see 6.2b).

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.3 Screening (JANTX, JANTXV, and JANS levels only). Screening shall be in accordance with table II of MIL-S-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

	Measurement	
Screen (see table II of MIL-S-19500)	JANS	JANTX and JANTXV
9	I_{CEX2} and h_{FE2}	Not applicable
11	I_{CEX2} and h_{FE2}; ΔI_{CEX2} = 100 percent of initial value or 1 nA dc, whichever is greater; Δh_{FE2} = 15 percent of initial value.	I_{CEX2} and h_{FE2}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CEX2} = 100 percent of initial value or 1 nA dc, whichever is greater; Δh_{FE2} = 25 percent of initial value.	Subgroup 2 of table I herein, ΔI_{CEX2} = 100 percent of initial value or 2 nA dc, whichever is greater; Δh_{FE2} = 25 percent of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: T_A = Room ambient in accordance with the general requirements of (see 4.5) of MIL-STD-750. V_{CB} = 10 V dc, P_T = 200 mW.

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4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Condition
B4	1037	V_{CB} = 10 V dc; P_T = 200 mW at T_A = 25°C; t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced-air cooling on devices shall be permitted.
B5	1027	T_A = 125°C ±25°C; for 96 hours; P_T = 200 mW at 100°C or adjusted as required by the chosen T_A to give an average lot T_J = 275°C. Marking legibility requirements shall not apply; V_{CB} = 10 V dc.

4.4.2.2 Group B inspection, table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500.

Subgroup	Method	Condition
B3	1027	P_T = 200 mW; T_A = +30°C ±5°C; V_{CE} = 10 V dc.
B5		Not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein.

Subgroup	Method	Condition
C2	2036	Test condition E.
C6	1026	P_T = 200 mW; T_A = 25°C ±3°C; V_{CE} ≥ 10 V dc.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in section 4 of MIL-STD-750.

4.5.2 Collector-base time constant. This parameter may be determined by applying an RF signal voltage of 0.5 volt (rms) across the collector-base terminals, and measuring the ac voltage drop (V_{eb}) with a high impedance RF voltmeter across the emitter-base terminals. With f = 31.8 MHz used for the 0.5 volt signal, the following computation applies: $r'_b C_c$: (ps) = 10 x V_{eb} (millivolts).

4.5.3 Base cutoff current. This parameter shall be measured using method 3041, bias condition A, of MIL-STD-750. The value of the base to emitter cutoff current shall be read on the current meter inserted in the base circuit.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 10 mA dc	$V_{(BR)CEO}$	15		V dc
Breakdown voltage collector to base	3001	Bias condition D; I_C = 10 µA dc	$V_{(BR)CBO}$	15		V dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 4.5 V dc	I_{EBO}		10	µA dc
Collector to emitter cutoff current	3041	Bias condition A; V_{CE} = 10 V dc; V_{EB} = 0.4 V dc	I_{CEX1}		50	nA dc
Collector to emitter cutoff current	3041	Bias condition A; V_{CE} = 10 V dc; V_{EB} = 2.0 V dc	I_{CEX2}		5.0	nA dc
Base cutoff current		V_{CE} = 10 V dc; V_{BE} = 2.0 mA dc See 4.5.3	I_{BEX}		5.0	nA dc
Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 1.0 mA dc	h_{FE1}	25		
Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 10 mA dc; pulsed (see 4.5.1)	h_{FE2}	30	150	
Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 30 mA dc; pulsed (see 4.5.1)	h_{FE3}	20		
Base to emitter voltage (non-saturated)	3066	Test condition B; V_{CE} = 1.0 V dc; I_C = 1.0 mA dc	V_{BE1}		0.8	V dc

See footnote at the end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Base to emitter voltage (nonsaturated)	3066	Test condition B; V_{CE} = 1.0 V dc; I_C = 10 mA dc	V_{BE2}		1.0	V dc
Saturation voltage and resistance (collector-emitter)	3071	I_C = 1.0 mA dc; I_B = 0.1 mA dc	$V_{CE(sat)1}$		0.15	V dc
Saturation voltage and resistance (collector-emitter)	3071	I_C = 10 mA dc; I_B = 1.0 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$		0.35	V dc
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition A; V_{CE} = 10 V dc; V_{BE} = 2.0 V dc	I_{CEX3}		5.0	µA dc
Low temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 10 mA dc; pulsed (see 4.5.1)	h_{FE4}	15		
Subgroup 4						
Magnitude of common-emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 4.0 V dc; I_C = 5.0 mA dc; f = 100 MHz	$\lvert h_{fe1} \rvert$	15		
Magnitude of common-emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 10 mA dc; f = 100 MHz	$\lvert h_{fe2} \rvert$	20		
Open circuit output capacitance	3236	V_{CB} = 4.0 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		2.5	pF
Input capacitance (output open-circuited)	3240	V_{EB} = 0.5 V dc; I_C = 0; 100 kHz ≤ f ≤ 1 MHz; See 4.5.1	C_{ibo}		2.5	pF

See footnote at end of table.

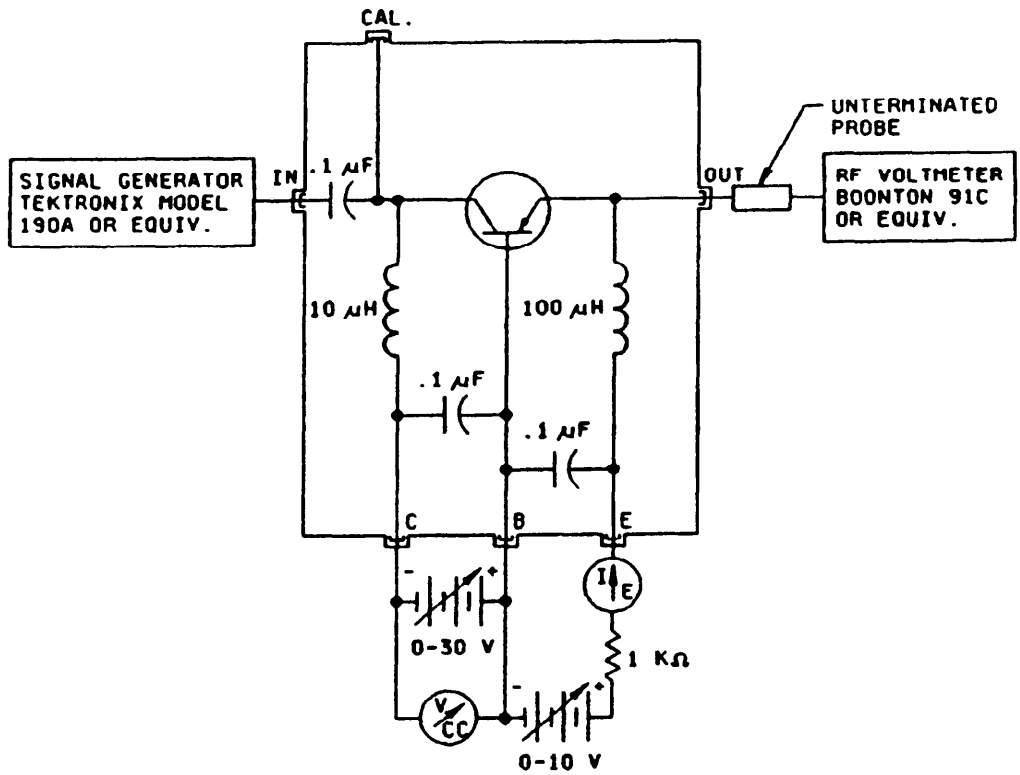
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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Collector-base time constant		V_{CE} = 4.0 V dc; I_C = 5.0 mA dc; f = 31.8 MHz; See 4.5.2 and figure 3	$r'_b C_{c1}$		60	ps
Collector-base time constant		V_{CE} = 4.0 V dc; I_C = 10 mA dc; f = 31.8 MHz; See 4.5.2 and figure 3	$r'_b C_{c2}$		50	ps
Pulse response	3251	Test condition B; except test circuit and pulse requirements per figure 4 herein. V_{CC} = 17 V dc; I_C = 10 mA dc	t_{on} t_{off}		2.5 3.5	ns ns
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan, see MIL-S-19500.

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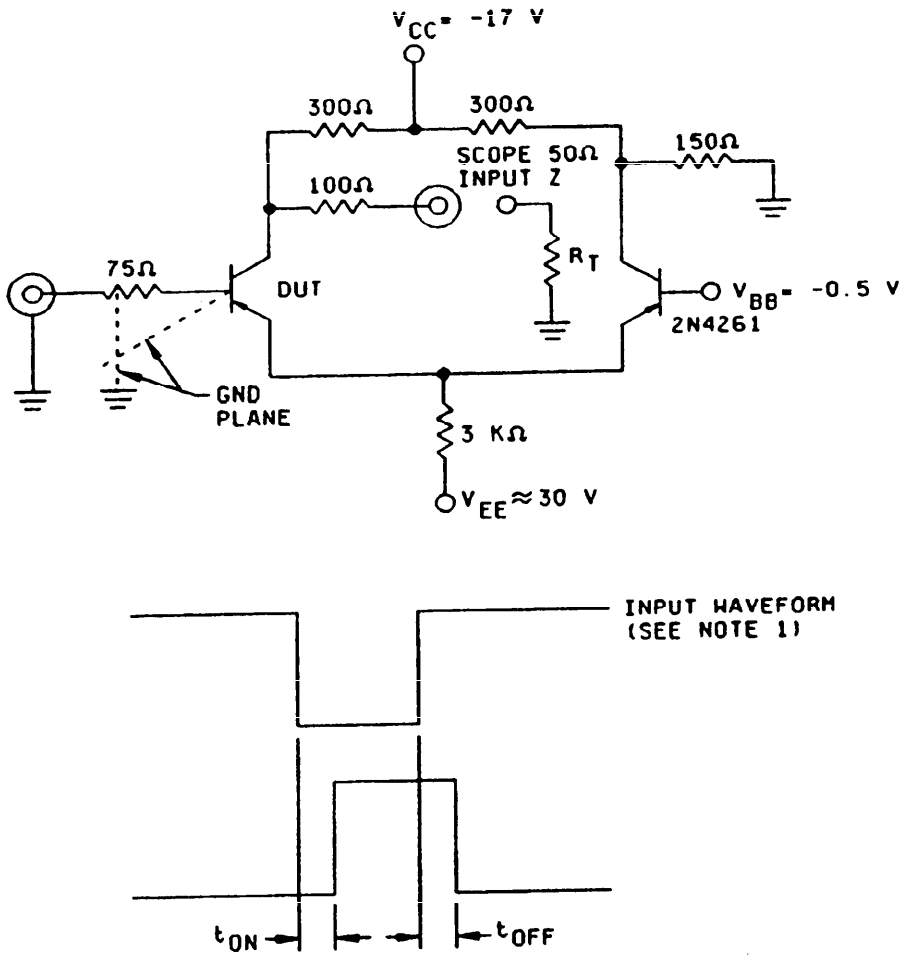
Procedure:

1. Set signal generator to 31.8 MHz and connect to "in" connector on test jig.
2. Connect low voltage dc power supplies as shown. A 1 KΩ resistor should be placed in series with the emitter power supply to prevent damage to transistors being tested.
3. Place transistor to be tested in socket. Set collector supply for V_{CE} = 4.0 V dc, and emitter supply for I_C as specified.
4. Connect RF voltmeter with unterminated probe adapter to "CAL" connector on test jig. Adjust signal generator until RF voltage is .5 volts. (Note: Decade switching of voltmeter should be accurate from 1 mV to 3 volts. If not, input voltage may be set using voltage dividers, utilizing lower scales of the RF voltmeter. If this done, the voltage dividers should be left in place when the voltmeter is removed, as they constitute a load on the input of the circuit.
5. Remove RF voltmeter from "CAL" connector and connect to "OUT" connector. Meter will now read r'_b, C_c as follows:

Meter range full scale	r'_b, C_c range
.003 volts	10 to 30 ps
.01 volts	30 to 100 ps
.03 volts	100 to 300 ps

FIGURE 3. Collector-base time constant test circuit (an equivalent circuit may be used).

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NOTES:
1. The input waveform is supplied by a pulse generator with the following characteristics: Z_{OUT} = 50 Ω; PW = 30 ns; duty cycle ≤ 2 percent.
2. Resistors shall be noninductive types.
3. The dc power supplies may require additional by-passing in order to minimize ringing.

FIGURE 4. Switching time test circuit and waveforms.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish (see 3.3.1).

c. Product assurance level and type designator.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodian:
Air Force - 17
NASA - NA

Users:
Air Force - 13, 15, 19, 85, 99

Preparing activity:
DLA - ES

(Project 5961-1486)

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INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/511B	2. DOCUMENT DATE (YYMMDD) 94/06/24

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, SWITCHING TYPE 2N4261, JAN, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

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c. ADDRESS (Include Zip Code) Defense Electronics Supply Center Attn: DESC-ELDT Dayton, Ohio 45444-5765	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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